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CLIENT PROJECT PROPOSAL

AI-Powered Academic Admission Assistance, Visitor Journey & Queue Management System

AI + WhatsApp + Smart Queue + Multi-Stage Admission Workflow

Prepared for: College / University Client

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1. Project Overview

During admission season, parents and students arrive at college with little knowledge of the process. Common difficulties include:

- **Whom to meet** — it is often unclear which staff member or officer handles a requirement.
- **Which department** — visitors may not know the right department or programme.
- **Multiple stages** — admission involves counselling, verification, accounts and registration counters.
- **Physical queues** — long waits without any indication of position or time.
- **Repeated questions** — reception staff answer the same questions again and again.
- **Finding rooms and staff** — campus layouts are unfamiliar to first-time visitors.
- **What happens next** — visitors often leave one counter unsure of the next step.

This proposal presents one integrated platform that addresses all of these: a **QR-enabled, WhatsApp-connected, AI-assisted visitor journey and queue management system**. It combines a receptionist registration desk, a permanent Visitor ID, real digital queues, configurable multi-stage workflows and automatic next-stage allocation with WhatsApp notifications, so that every visitor is guided smoothly and every staff member works from a clear, live queue.

2. Core Concept

The system is **not simply a chatbot**. It is a **digital visitor journey, queue and workflow management system with an AI admission assistant**. The assistant is one part of the platform; the queues and workflows are the engine.

Every visitor is represented by **one permanent Visitor ID** for the whole visit:

```
V121 — Raj Kumar — Purpose: Admission — Department: ECE
    → Counselling → Eligibility Verification → Document Verification
    → Accounts → Admission Confirmation → Registration → Completed
```

The same Visitor ID follows the visitor through every stage, even though they receive a fresh queue token at each stage. This gives the college complete traceability, a clean audit trail and accurate statistics for every admission.

3. Complete Visitor Journey

```
Parent arrives → QR scan → WhatsApp → AI Assistant
    → Receptionist registration → Visitor ID → Department identification
    → Initial queue → Staff processing → Stage completion
    → Automatic next stage → Next queue → WhatsApp notification
    → Next staff / location → Final completion
```

1. **Arrival:** the parent or student walks into the college.
2. **QR scan:** scanning the QR opens the college's WhatsApp assistant (Section 4).
3. **AI conversation:** the AI answers early questions and explains the process.
4. **Receptionist registration:** the visitor is formally registered and receives a Visitor ID.
5. **Department identification:** a known department maps to its workflow; an unsure visitor goes to the help desk.
6. **Queue:** the visitor is placed in the correct first-stage queue with a token.
7. **Staff processing:** the assigned staff member serves stages like Counselling or Document Verification.
8. **Automatic next stage:** on completion, the system assigns the next stage and queue automatically.
9. **WhatsApp notification:** the visitor is told exactly where to go and whom to meet.
10. **Final completion:** after the last stage the visitor status becomes Completed.

4. QR Code Entry

The QR code is the visitor's simplest gateway into the system. QR codes will be placed at the college **entrance**, the **reception**, the **admission block**, the **help desk** and on posters and signboards.

```
QR scan → Opens the college's WhatsApp assistant → Visitor starts chatting
```

Scanning the QR opens WhatsApp with the college's admission assistant. No separate mobile application is required for the visitor — anyone with WhatsApp on their phone is connected in seconds, with no number to remember and nothing to install.

5. WhatsApp Communication

Communication between the parent and the college runs over real WhatsApp through the official **Meta WhatsApp Business Platform / Cloud API**.

```
Parent WhatsApp → Meta Cloud API → Webhook → Backend
    → AI / Queue / Workflow → Backend → WhatsApp → Parent
```

Parents receive from WhatsApp:

- AI responses to admission and college questions,
- live queue updates and positions,
- turn-call notifications when it is their turn,
- next-stage information after each stage,
- room / location instructions, and
- general process updates throughout the visit.

Because parents commonly use WhatsApp daily, there is no learning curve for the visitor and no parent-facing application to develop or maintain.

6. AI Admission Assistant

The AI assistant runs on **Ollama** with a **local large language model**, running on the college's own hardware. It understands natural questions and answers in simple language.

```
Parent message → WhatsApp → Backend → AI Service → Ollama
    → Local LLM → Response → WhatsApp → Parent
```

Typical questions the assistant answers:

- "What documents do I need?", "Where is the admission office?",
- "What is the fee?", "What courses are available?",
- "Where should I go?", "What do I do next?"

Component	Responsibility
AI	Understands questions and communicates answers.
Database	Stores authoritative college information.
Workflow engine	Determines the visitor's actual process.
Queue engine	Determines the real queue state and positions.
Staff	Control and complete real-world admission stages.

Important: the AI is strictly **read-only** regarding queues and workflows. It can report, explain and guide, but it cannot independently change a queue position or a workflow stage. Only authorized staff and receptionist actions change official state.

7. College Knowledge Base

The AI answers from a **college-managed knowledge base** rather than from imagination. The knowledge base holds college details, departments, courses, fees, eligibility, required documents, hostel,

transport, timings, locations and frequently asked questions.

Parent question	Sourced from
"What documents do I need?"	Departmental / admission FAQ
"What is the fee?"	Fees entry in the knowledge base
"Is hostel available?"	Hostel entry
"What are the timings?"	Timings entry
"Where is the admission office?"	Location entry
"What is my queue position?"	Live queue data (not the knowledge base)

The knowledge base is maintained through the Admin dashboard, so answers always reflect the college's own current information.

8. Counselling / Help Desk Routing

When a parent does not know which department they need, the system **never randomly selects one**.

> **Parent:** "I don't know which department I need."
> **AI:** "No problem — our counselling / help desk team will guide you based on your interests and marks. You will not be assigned to any department automatically."

- The visitor is registered under **"Needs Counselling / Help Desk"**.
- The system routes them into the dedicated **Counselling / Help Desk** workflow and queue.
- Help desk staff interview the visitor and direct them to the correct department and process.

This prevents incorrect routing, wasted staff time and a confusing journey for the visitor.

9. Receptionist System

The receptionist remains the human point of contact connecting the physical visitor to the digital journey. Registration takes seconds:

```
Name: Raj Kumar      Mobile / WhatsApp: 9876543210
Purpose: Admission   Department: ECE (or "Needs Counselling / Help Desk")
Remarks: Walk-in parent            →  [Register Visitor]  →  Visitor ID: V121
```

On registration the system creates the Visitor ID, selects the correct workflow, places the visitor in the right first-stage queue, records the stage history and sends a welcome WhatsApp message with the Visitor ID, stage, queue token and instructions.

10. Digital Queue Management

The system replaces physical queues with **real digital queues** backed by the database. Every workflow stage can have its own separate queue. Each entry carries a queue number, current position and live status.

Status	Meaning
WAITING	Visitor is in the queue waiting for their turn.
CALLED	Staff has announced the visitor's number.
SERVING	Staff is currently attending to the visitor.
COMPLETED	The stage is completed (visitor may auto-move to the next queue).
HOLD	Visitors processing is paused; others are served first.
SKIPPED	Visitor left this queue run (e.g. rescheduled).
CANCELLED	Visitor record cancelled (no-show / left college).

Staff and receptionist can see the queue number, live position and people ahead; call the next visitor; hold or skip a visitor with a recorded reason; reassign a visitor to another stage; and complete a stage. Visitors get their real position on WhatsApp — no more guessing.

11. Multi-Department Support

The system serves the whole college across departments such as **CSE, ECE, EEE, Mechanical, MBA** and the shared **Help Desk**. Each department maintains its own queues, staff and process.

The platform deliberately does **not** assume every department follows the same procedure. Each department can define its own workflow stages, order, staff, rooms and queues (Section 12).

12. Multi-Stage Admission Workflow

The **workflow engine** models each department's admission process as a configurable chain of stages. For example:

```
ECE Admission:  Counselling → Eligibility Verification → Document Verification
                → Accounts → Admission Confirmation → Registration
CSE / EEE / MECH / MBA: separate configurable chains
Counselling / Help Desk: a single help-desk stage for unsure visitors
```

For each workflow the administrator can configure the department, the list and **order of stages**, the staff member, the room/counter and the queue for every stage, and whether a stage is active. Workflows are configuration data — not hard-coded logic — so the admission process can be changed from the Admin dashboard without rebuilding the system.

13. Automatic Next-Stage Allocation

When staff complete the current stage, the system automatically performs seven actions:

```
Staff clicks COMPLETE
  1. Marks the stage complete
  2. Finds the next workflow stage
  3. Identifies the appropriate queue
  4. Adds the visitor to that queue
  5. Updates the visitor's current stage
  6. Identifies the next staff / location
  7. Sends the WhatsApp notification
```

Example: Counselling completed → the visitor is added to the Document Verification queue and receives: "*Counselling completed. Your next step is Document Verification. Please wait for your turn.*"

The same chain repeats automatically until the entire workflow is completed and the visitor's status becomes Completed.

14. Staff Dashboard

Each staff member sees only the queues they are assigned to, showing the current queue, waiting visitors, the visitor currently being served, visitor details, the stage, the room and the counter.

Action	What it does
CALL NEXT	Summons the earliest waiting visitor and notifies them on WhatsApp.
HOLD	Pauses the visitor (reason recorded) so others are served first.
SKIP	Removes the visitor from this queue run (reason recorded).
REASSIGN	Moves the visitor to another stage / queue.
VIEW	Opens the visitor's full profile and history.
COMPLETE	Finishes the stage and triggers automatic next-stage allocation.

The dashboard is focused and role-aware: a staff member never sees queues that do not belong to them.

15. Admin Dashboard

The Admin dashboard gives the college complete control of its configuration:

- departments, staff, rooms and counters,
- workflows and workflow stages (order, staff, room, queue, active status),
- FAQs and the college knowledge base,
- college information (fees, hostel, transport, timings, locations),
- queue settings and visitor records.

This is what makes the system **configurable rather than hard-coded**: technology adapts to the college's process, and most changes are made from the dashboard without any development work.

16. Database

The system uses **SQLite** with the **SQLAlchemy** object-relational layer. A database is essential so that every visitor, queue, workflow, notification and message is stored reliably and queryable in real time.

Major entities:

- **Users · Staff · Departments · Rooms** — who logs in, who works where, and the college structure.
- **Visitors** — every registered visitor with a permanent Visitor ID.
- **Workflows · Workflow Stages** — each department's ordered admission process.
- **Queues · Queue Entries** — one real queue per stage plus each visitor's token and status.
- **Visitor Stage History** — the complete audit trail of each journey.
- **FAQs · College Information** — the knowledge base used by the AI.
- **Notifications · Chat Messages** — the communication log.

Relationships such as *Visitor → Queue Entry → Queue → Stage → Workflow → Department* let the system answer questions like "what is V121's full journey?", "who is ahead in this queue?" and "how many visitors are waiting per stage" instantly.

17. Backend

The backend uses **Python + FastAPI** and acts as the central nervous system. Its responsibilities include:

- authentication and role enforcement,
- visitor management (registration, search, status, history),
- queue management (tokens, positions, call-next),
- workflow management (stages, order, next-stage logic),
- staff actions (call, hold, skip, reassign, complete),
- AI communication with Ollama,
- the WhatsApp webhook,
- notification generation, and
- all database operations.

FastAPI is modern, fast and well documented, and cleanly separates routes, business logic and data access.

18. Frontend

The frontend uses **React**, bundled by **Vite** and styled with **Tailwind CSS**. It provides the complete web interface used by the college team:

- Admin, Receptionist and Staff dashboards,
- visitor registration and queue management screens,
- workflow management screens,
- forms, tables, live status indicators and search/filtering,
- a responsive interface suitable for desktop, tablet and phone.

The interface is role-aware: each user sees only the menus and actions permitted by their role.

19. Authentication & Security

The system is protected with **JWT authentication**, **password hashing** and **role-based access control**.

Role	Permissions
ADMIN	Full control: configuration, knowledge base, statistics, all records.
RECEPTIONIST	Visitor registration, search, queues, reassignment, cancellation.
STAFF	Only assigned queues/stages and processing actions.

All credentials — WhatsApp tokens, secrets, the JWT signing key — are protected in environment variables and are never committed or hard-coded. The secure login flow, per-role permissions and HTTPS deployment keep college data safe.

20. Webhook Communication

A **webhook** is a secure callback: when a parent sends a WhatsApp message, Meta calls the college backend's webhook URL to deliver it, and the backend replies.

```
Incoming:  WhatsApp → Meta → Webhook → FastAPI
Outgoing:  FastAPI → Meta WhatsApp Cloud API → WhatsApp → Parent
```

Webhooks make the assistant genuinely real-time — the system is informed the instant a message arrives and can respond immediately, without polling.

21. Notification System

The system automatically sends WhatsApp notifications at key moments:

- **Registration confirmation** — Visitor ID, stage, token and instructions,
- **Queue status** — live position and people ahead on request,
- **Turn notification** — "Your turn is now. Please proceed to room…",
- **Next-stage notification** — automatic after each stage,
- **Room / location** and **staff instructions** — where and whom to meet,
- **Completion** — stage completion and final admission completion.

Notifications are generated at the moment of the real event, so they always reflect the true official state of the visit.

22. Complete End-to-End Example

Parent: Raj Kumar · **Visitor ID:** V121 · **Department:** ECE

```
Arrival → Reception → V121 created → Counselling queue
   → (AI chats & answers questions)
   → Staff CALLS V121 → Counselling COMPLETED
   → Document Verification queue → COMPLETED
   → Accounts queue → COMPLETED → Admission Confirmation → Registration
   → COMPLETED
```

Sample WhatsApp messages along the journey:

> **WhatsApp:** "Welcome Raj! Your Visitor ID is V121. You are in the Counselling queue. Number: ECE-COUNS-124."

> **WhatsApp:** "Your turn is now, ECE-COUNS-124. Please proceed to Counselling Room (R101). Meet: Mrs. Priya."

> **WhatsApp:** "Counselling completed. Your next step is Document Verification. You have been added to that queue. Please wait."

> **WhatsApp:** "All your admission stages are complete. Congratulations!"

23. System Architecture

```
Parent → QR → WhatsApp → Meta Cloud API → Webhook
    → FastAPI Backend → AI Service → Ollama → Local LLM
    → Database (SQLite) · Queue Engine · Workflow Engine
    → Admin / Receptionist / Staff dashboards → WhatsApp → Parent
```

The parent communicates through WhatsApp; Meta forwards messages to the backend webhook. The backend uses the AI service (Ollama plus the college knowledge base and the visitor's live state) to answer, and replies via the WhatsApp API. All visitors, queues, workflows and messages are stored in the database. College staff control and configure everything through role-based dashboards. In short: *WhatsApp talks to the parent, the backend orchestrates the journey, the database records everything, and staff stay in control.*

24. Technology Stack

Layer	Technology	Purpose
WhatsApp	Meta WhatsApp Business Platform / Cloud API	Real parent messaging, inbound and outbound
Backend	Python + FastAPI	Central API, webhook and orchestration
AI	Ollama + local LLM	Conversational admission assistant
Database	SQLite + SQLAlchemy	Persistent relational storage
Frontend	React + Vite	Admin, Receptionist and Staff dashboards
UI styling	Tailwind CSS	Modern responsive interface
Authentication	JWT + password hashing	Secure login and sessions
Communication	Webhook	Real-time inbound WhatsApp messages
Queue	Backend queue management	Real queues, positions, call-next
Workflow	Configurable workflow engine	Multi-stage admission processes
QR	QR-based WhatsApp access	Instant visitor entry point

25. Benefits to the College

Benefit	Impact
Less visitor confusion	Visitors always know whom to meet and where to go.
Reduced queue congestion	Visitors wait comfortably and come when called.
Better reception efficiency	Fast digital registration replaces paper and repetitive answers.
Better staff coordination	Everyone sees exactly their queue and workload.
Clear admission journey	The full path is visible to staff, admin and visitor.
Automated communication	WhatsApp updates remove the need to ask repeatedly.
Real-time queue information	Live positions and waiting lists for everyone.
Department-specific workflows	Each department follows its own correct process.
Reduced repetitive questions	The AI answers routine queries on WhatsApp.
Centralized records	Complete digital audit of every visitor journey.

26. Customization

The system is designed to be configured to the client's college without programming. Departments, courses, staff, rooms, counters, admission stages, FAQs, college policies and workflow rules are all configuration data. The implementation team will work with the college during setup to match the system exactly to its real admission process — for example its departments, its stage order and its counters.

27. Future Enhancements

The following capabilities may be added after the core system goes live. They are listed as **future enhancements** and are not part of the core platform initially:

- multilingual AI and multilingual WhatsApp support,
- voice-based assistance,
- admission document upload through WhatsApp,
- appointment scheduling,
- admission analytics and trend reports,
- a mobile staff application,
- cloud deployment,
- advanced document retrieval (RAG), and
- integration with other college systems (ERP, fee, student records).

28. Final System Summary

```
QR → WhatsApp → AI → Reception → Visitor ID → Queue → Staff → Workflow
   → Automatic Next Stage → Next Queue → WhatsApp Notification
   → Final Completion
```

The proposed system creates **one unified digital visitor journey** for every parent and student who visits the college for admission. From the QR scan at the entrance to the final admission confirmation, the visitor is welcomed, registered, guided, queued, called, served and informed — through one system, one WhatsApp conversation and one permanent Visitor ID. The college gains a calmer admission process, informed visitors, organized staff and complete digital records.

> The proposal describes a complete, integrated solution: an AI-powered admission assistant, real WhatsApp communication, QR-based access, digital queues and configurable multi-stage workflows that adapt to the college's exact admission process.

— End of proposal —